

|               |                  |              |              |              |              |              |              |
|---------------|------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Baichuan-13B  | BeamSearch       | 41.06        | 9.59         | 32.24        | 9.09         | 36.18        | 13.18        |
|               | GCN              | 57.67        | 31.47        | 55.51        | 30.16        | 62.11        | 37.05        |
|               | GAT              | 57.74        | 31.87        | 54.95        | 29.24        | 62.11        | 37.05        |
|               | GraphSAGE        | <b>59.32</b> | <b>34.36</b> | <b>56.15</b> | <b>31.60</b> | <b>65.18</b> | <b>40.49</b> |
|               | GIN              | 57.38        | 31.17        | 55.08        | 30.04        | 62.11        | 37.05        |
|               | GraphTransformer | 59.15        | 34.36        | 56.06        | 31.12        | 64.52        | 40.14        |
| Vicuna-13B    | Direct           | 50.46        | 21.27        | 53.57        | 23.19        | 73.70        | 45.80        |
|               | BeamSearch       | 56.64        | 26.93        | 54.09        | 26.19        | 54.55        | 23.60        |
|               | GCN              | 59.46        | 33.14        | 62.48        | 38.89        | 83.05        | 62.95        |
|               | GAT              | 59.28        | 33.39        | 62.96        | 39.24        | 83.05        | 62.95        |
|               | GraphSAGE        | <b>61.86</b> | <b>35.68</b> | <b>63.71</b> | <b>39.88</b> | <b>86.07</b> | <b>67.63</b> |
|               | GIN              | 59.14        | 32.33        | 62.61        | 38.82        | 83.05        | 62.95        |
|               | GraphTransformer | 59.57        | 33.47        | 63.32        | 39.38        | 85.41        | 66.28        |
| CodeLlama-7B  | Direct           | 58.06        | 29.39        | 59.44        | 30.83        | 84.12        | 62.89        |
|               | BeamSearch       | 60.34        | 31.36        | 64.12        | 38.99        | 83.25        | 63.48        |
|               | GCN              | 65.07        | 40.50        | 67.46        | 45.84        | 87.23        | 69.27        |
|               | GAT              | 65.20        | 40.93        | 67.41        | 46.46        | 87.23        | 69.27        |
|               | GraphSAGE        | <b>66.67</b> | <b>43.03</b> | 67.97        | 46.31        | <b>88.53</b> | <b>72.02</b> |
|               | GIN              | 65.52        | 40.98        | 66.89        | 45.41        | 87.23        | 69.27        |
|               | GraphTransformer | 65.83        | 42.58        | <b>68.90</b> | <b>47.20</b> | 88.36        | 71.72        |
| Mistral-7B    | Direct           | 60.60        | 30.23        | 69.83        | 39.85        | 84.26        | 53.63        |
|               | BeamSearch       | 67.13        | 36.73        | 73.55        | 47.12        | 85.87        | 61.53        |
|               | GCN              | 66.54        | 40.74        | 73.34        | 50.76        | 86.39        | 65.49        |
|               | GAT              | 66.77        | 40.74        | 73.36        | 50.20        | 86.39        | 65.49        |
|               | GraphSAGE        | 68.12        | <b>43.09</b> | <b>75.51</b> | <b>52.94</b> | 87.51        | 66.57        |
|               | GIN              | 66.69        | 40.79        | 72.89        | 50.44        | 86.39        | 65.49        |
|               | GraphTransformer | <b>68.26</b> | 43.08        | 73.80        | 51.45        | <b>88.25</b> | <b>67.84</b> |
| CodeLlama-13B | Direct           | 57.55        | 28.88        | 68.57        | 41.79        | 91.20        | 76.07        |
|               | BeamSearch       | 62.65        | 34.31        | 69.53        | 43.35        | 91.74        | 76.60        |
|               | GCN              | 66.22        | 41.05        | 72.99        | 52.18        | 91.83        | 77.88        |
|               | GAT              | 66.29        | 41.28        | 74.08        | 53.56        | 91.83        | 77.88        |
|               | GraphSAGE        | <b>67.30</b> | <b>42.41</b> | <b>74.93</b> | <b>54.52</b> | <b>93.84</b> | 80.38        |
|               | GIN              | 66.40        | 40.89        | 73.62        | 53.15        | 91.83        | 77.88        |
|               | GraphTransformer | 66.70        | 42.07        | 74.72        | 54.10        | 93.81        | <b>80.44</b> |
| GPT-3.5-turbo | Direct           | 73.85        | 45.73        | 82.85        | 62.07        | 96.09        | 83.65        |
|               | BeamSearch       | 75.51        | 49.62        | 83.57        | 64.50        | 95.66        | 82.72        |
|               | GCN              | 76.93        | 51.43        | 84.92        | 65.05        | 96.38        | 86.15        |
|               | GAT              | 75.63        | 49.36        | 84.77        | 65.48        | 96.38        | 86.15        |
|               | GraphSAGE        | <b>77.90</b> | <b>52.68</b> | <b>85.29</b> | <b>65.80</b> | <b>96.43</b> | <b>86.26</b> |
|               | GIN              | 76.86        | 51.00        | 84.14        | 64.30        | 96.38        | 86.15        |
|               | GraphTransformer | 77.61        | 52.30        | 84.21        | 64.32        | 96.38        | 86.19        |
| GPT-4-turbo   | Direct           | 77.60        | 52.18        | 88.29        | 69.38        | 97.36        | 84.58        |
|               | BeamSearch       | 77.56        | <b>52.54</b> | 88.16        | <b>70.39</b> | 97.35        | <b>85.78</b> |
|               | GCN              | 77.01        | 50.49        | 88.56        | 69.60        | 97.10        | 85.22        |
|               | GAT              | 76.41        | 49.66        | 88.43        | 69.52        | 97.10        | 85.22        |
|               | GraphSAGE        | <b>78.76</b> | 52.53        | <b>88.63</b> | 69.65        | 97.34        | 85.67        |
|               | GIN              | 77.74        | 51.02        | 88.05        | 69.13        | <b>97.36</b> | 84.58        |
|               | GraphTransformer | 78.47        | 52.17        | 88.07        | 68.71        | 97.32        | 85.57        |